

IMPERIAL

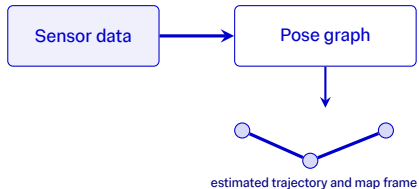
Sonar-Based Underwater SLAM with 3D Gaussian Splatting Mapping

Saxon Shang supervised by Dr. Sen Wang
June 24th 2026

SLAM and 3D Gaussian Splatting Mapping

Two complementary building blocks

Simultaneous Localisation and Mapping

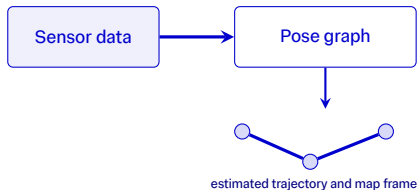


- Estimates robot trajectory while building or using a map.
- Fuses motion and observations to reduce accumulated drift.

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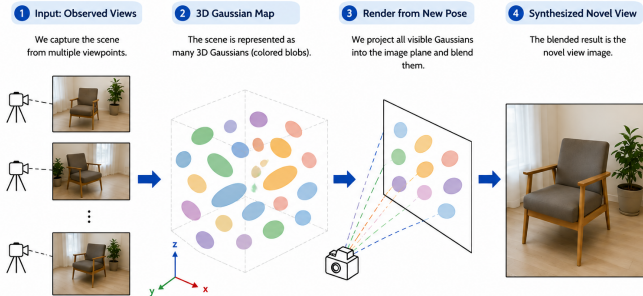
Simultaneous Localisation and Mapping



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3D Gaussian Splatting Mapping

3D Gaussian Splatting: Novel View Synthesis



Why Underwater Sonar SLAM Is Hard

Can a renderable sonar map provide online correction factors?

1. Problem setting

Underwater robots cannot use GNSS. IMU/DVL odometry therefore drifts over time.

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6. Project idea

Use render consistency between live sonar and 3D Gaussian Splatting prediction as a SLAM constraint.

Research Gap: Sonar SLAM + 3DGS

Underwater SLAM

- Fuses sonar, camera, IMU, and DVL.
- Uses feature, image, or sensor factors.

Li et al. 2018; Cardaillac and Ludvigsen 2023; Xu et al. 2025.

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3DGS SLAM

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- Current systems are visual or LiDAR-based.

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Underwater 3DGS

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- Focuses on reconstruction and rendering, not SLAM.

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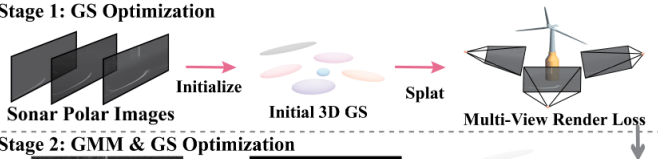
Missing integration in the reviewed literature

Live sonar → sonar 3DGS candidate render → gated Pose3 factor → iSAM2.

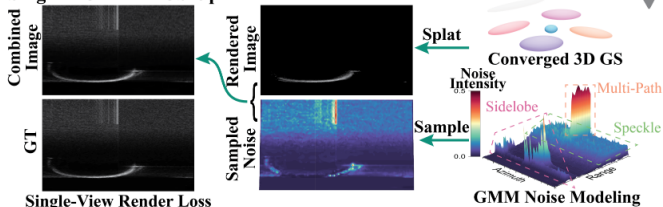
NAS-GS: Noise-Aware Sonar Gaussian Splatting

A renderable sonar measurement model reference

Stage 1: GS Optimization



Stage 2: GMM & GS Optimization



NAS-GS training pipeline, adapted from Xu et al.

What NAS-GS does

- Learns a 3D Gaussian sonar scene from polar sonar images.
- Optimises rendered structure together with acoustic noise appearance.
- Produces a differentiable sonar renderer for new viewpoints.

How I use it

- Freeze the trained map during replay.
- Render candidate Pose3 hypotheses online.
- Insert a factor only when live sonar and render agree.

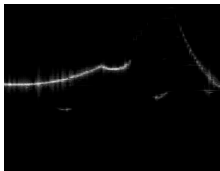
SLAM Pose graph optimization with Pose3 constraints and iSAM2 backend

NAS-GS predicts observations; iSAM2 consumes accepted constraints

Online loop: render from the map, compare with the live frame, then add a graph factor only when the evidence is strong.

Renderable sonar map

NAS-GS predicts what the sonar should observe from a proposed Pose3 camera pose.



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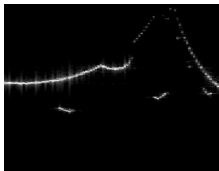
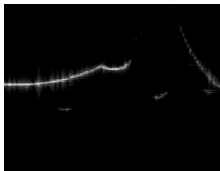
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Online evidence

A live sonar frame is compared against rendered candidates around the current drifted pose.



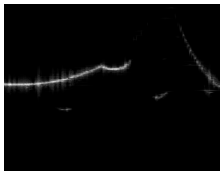
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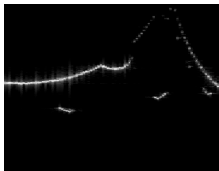
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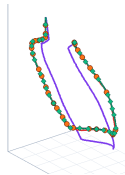
Online evidence

A live sonar frame is compared against rendered candidates around the current drifted pose.



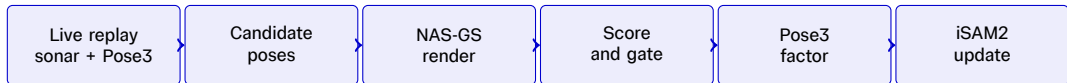
Graph correction

If render similarity and gates pass, the recovered pose is inserted as a Pose3 factor.



Online-render Registration Backbone

Same online loop for same-route correction and revisit correction



Local single

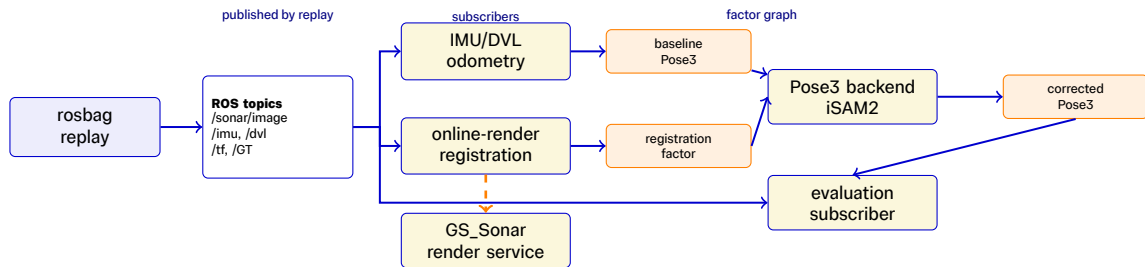
Compact x, y, z, ψ search around the live Pose3 baseline for nearby pose correction.

Route-progress revisit

Ordered sequence render search guided by route progress and drift-prior checks.

Gating is configured separately from the search mode. Score and structure checks prevent weak render matches from entering the graph.

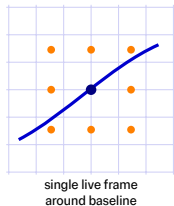
Algorithm Step 1: Live Replay System



The replay is deliberately online. Nodes subscribe to ROS topics in time order, the registration node queries NAS-GS only at candidate poses, and iSAM2 receives only gated Pose3 factors.

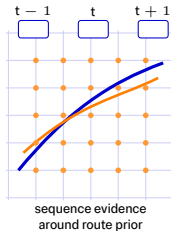
Algorithm Step 2: Candidate Pose And NAS-GS Render

Local single-frame search



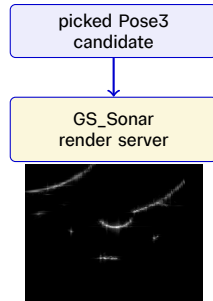
Compact x, y, z, ψ grid for same-route correction.

Route-progress sequence search



The orange line is a smooth candidate correction sequence. Wider candidates are allowed, but neighbouring frames must agree.

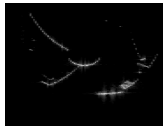
Render query



The output is a rendered sonar image for the proposed robot pose.

Algorithm Step 3: Score And Gate

Accepted pair



live sonar



rendered sonar

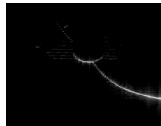
Frame 24 selects rendered frame 24 and passes the score gate.

score gate	$s = 0.926 \geq 0.08$, pass
pose gate	shift = 0.048 m, pass
replay evidence	published as a Pose3 registration factor

Score calculation

Intensity normalisation, NCC-style similarity, residual penalty, and bright-return overlap are combined into one render score. Minimum threshold $s_{\min} = 0.08$.

Rejected pair



live sonar



rendered sonar

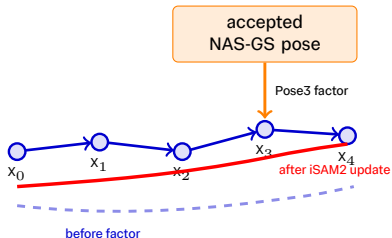
Frame 228 selects rendered frame 216 but falls just below threshold.

score gate	$s = 0.0799 < 0.08$, fail
edge check	best candidate on search edge
replay evidence	one of 6 score-below-threshold events

Gates before insertion

Render-score, structure-support, pose-jump, and temporal-consistency gates must pass before publishing a Pose3 factor to iSAM2.

Algorithm Step 4: Pose3 Factor And iSAM2 Update



Before

The graph contains IMU/DVL odometry factors, so drift accumulates along the chain.

Insertion

The accepted candidate pose and covariance are published as a Pose3 registration factor.

After

iSAM2 optimises the graph, pulling nearby states toward the render-consistent pose.

Algorithm Step 5: Route-progress Revisit Priors



Route prior

- Express the current revisit frame as normalised route progress from 0 to 1.
- Select the corresponding ordered window on the mapping route instead of trusting the nearest drifted pose.
- Use this window only as a proposal; sequence render evidence selects the final factor pose.

Drift prior

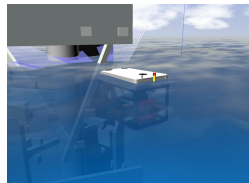
- Learn expected drift magnitude from aligned baseline-to-GT errors, using 21-sample smoothing and a PCHIP curve over travelled distance.
- Query absolute XY, z, and yaw magnitudes at the current distance, allowing transfer when route direction changes.
- Reject only excess beyond the learned envelope: 3.0 m XY, 2.5 m z, or 0.70 rad yaw. NAS-GS selects the pose inside it.

Simulation Platform: DAVE In Docker

Repeatable sonar, IMU, DVL, and ground-truth recording

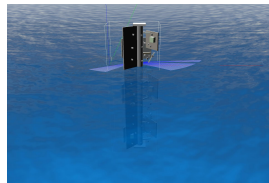
Docker setting and robot model

- CUDA-enabled Docker container.
- DAVE with ROS Noetic and Gazebo.
- BlueROV2 vehicle model.
- Sensors recorded: sonar, IMU, DVL, odometry, and pose ground truth.



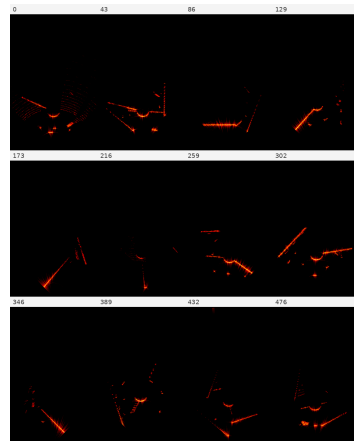
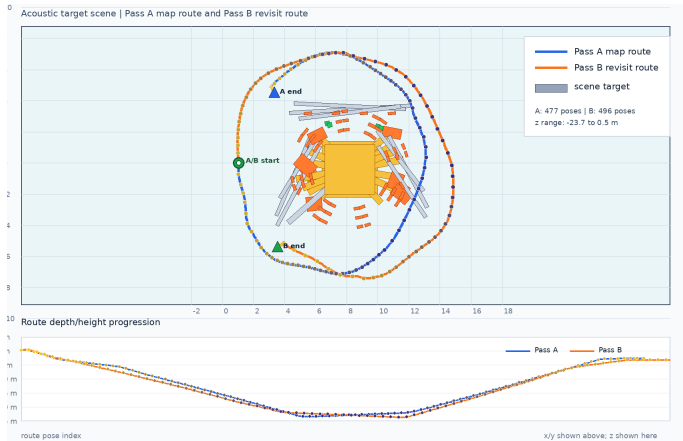
Custom scene description

- A fixed underwater target structure creates repeatable sonar returns.
- Pass A, Pass A revisit, and Pass B use controlled waypoint routes.
- Sonar field of view, range, beam pattern, and noise are explicitly set.



Simulation Path Recording

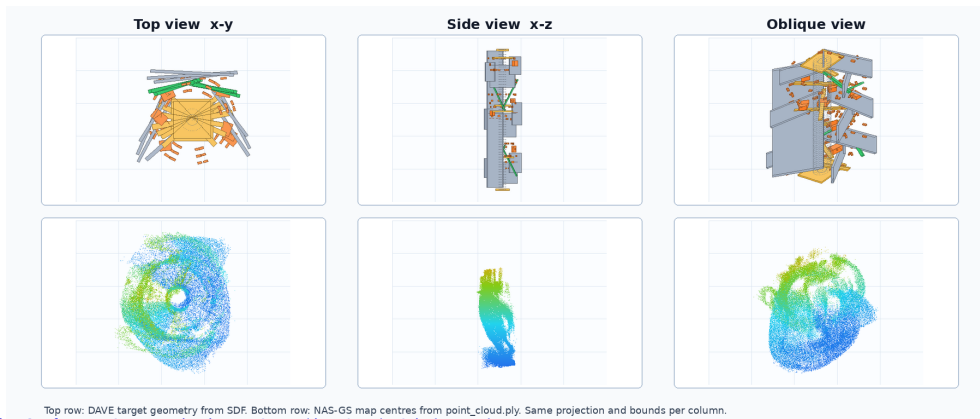
Pass A trains the map; Pass B reuses the start point but follows a hard revisit path



From Simulated Geometry To NAS-GS Map

The learned map

The two rows use the same top, side, and oblique projections. They differ because the DAVE row shows the full simulated target geometry, while the NAS-GS row shows Gaussian centres learned only from sonar-visible acoustic returns.



Top row: DAVE target geometry from SDF. Bottom row: NAS-GS map centres from point_cloud.ply. Same projection and bounds per column.

Experimental Design

Recorded data, map training, and replay policy

Case	Images	Sonar setting	NAS-GS map	Replay policy
Pass A	477	90°, 20 m	Pass A 900k/i1000, PSNR 35.77 dB	Local single
Pass A revisit	431	90°, 20 m	Pass A 900k/i1000, PSNR 35.77 dB	Local single; route-progress check
Pass B revisit	496	90°, 20 m	Pass A 900k/i1000, PSNR 35.77 dB	Route-progress sequence
Offshore	1139	120°, 12 m	Offshore 750k/i800, PSNR 33.68 dB	Local single

Baseline setting

IMU/DVL Pose3 replay with NAS-GS factors disabled. Noise is injected as DVL scale 1.10, body bias (0.030, -0.025, 0.0) m/s, yaw drift 8.0×10^{-4} rad/s, and 2.5 s DVL dropout every 35 s after 45 s.

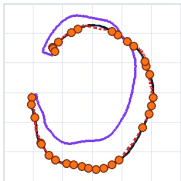
Pass A Policy Selection

The BP-rendered library with runtime no-BP scoring was selected

BP means beam-pattern rendering. It applies the sonar's angular transmit/receive sensitivity as an intensity weighting, attenuating returns away from the central beam. **no-BP** omits this directional modulation.

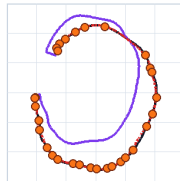
BP90x20 rendering

36 factors XY RMSE 2.741 to 0.733 m mean |z| 12.719 to 0.901 m



no-BP rendering

31 factors XY RMSE 2.741 to 0.612 m mean |z| 12.719 to 0.467 m



— GT sonar mount

— baseline

— corrected

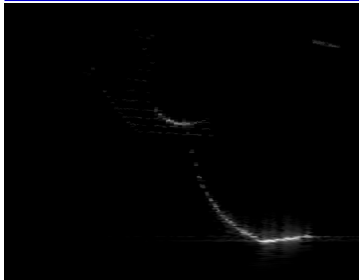
● accepted NAS-GS factor

Pass A Sonar Evidence

One recorded frame and the corresponding BP/no-BP rendered libraries

Recorded Pass A

Live sonar frame from the DAVE route.



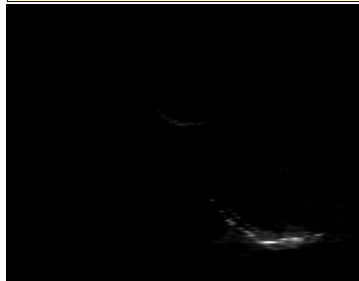
Rendered BP library

Reference image from the trained NAS-GS map.



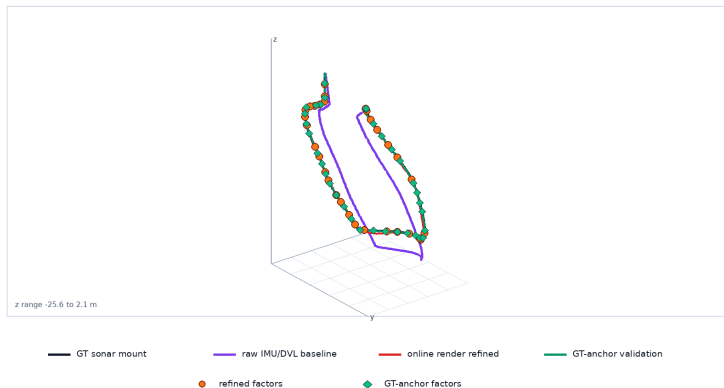
Rendered no-BP library

Alternative rendered appearance for policy testing.



Pass A Result

Pass A 3D z-drift diagnostic with map-pose registration



Metrics: 30 factors. XY RMSE $2.741 \rightarrow 0.570$ m; XYZ RMSE $15.078 \rightarrow 0.937$ m; mean $|z|$ $12.719 \rightarrow 0.569$ m.

Final error: XY 0.305 m; $|z| = 0.069$ m.

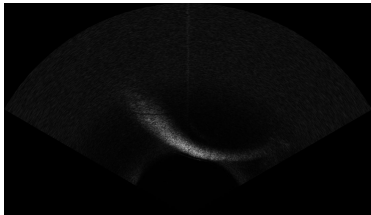
Offshore Dataset: Real Sonar Validation

Wind-turbine inspection sequence

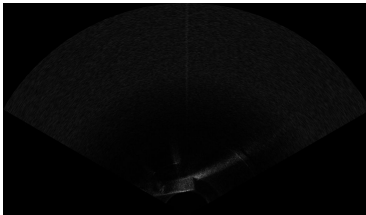
- Real sequence recorded around an offshore wind-turbine foundation with a Tritech Gemini 1200ik imaging sonar.
- More than 20 minutes and **1,139 frames** from the surface towards the seabed, using a 120° field of view and 12 m range.
- Sidelobe, speckle, multipath, and strong vertical motion make this a demanding real-acoustic transfer case.

Reference and validation role

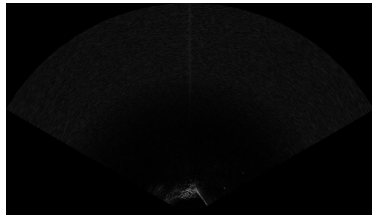
Aqua-SLAM 6-DoF poses supervise NAS-GS and provide the replay reference. This is not independent survey ground truth.



Frame 216: strong curved return



Frame 517: partial return



Frame 864: sparse near-range return

BP versus no-BP on Offshore Data

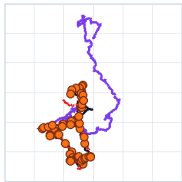
The better rendering mode depends on the recorded sonar appearance

Offshore online-render with BP and no-BP rendering

Same relaxed local single policy on real sonar with different NAS-GS render settings

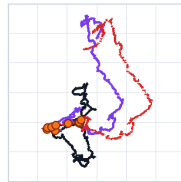
BP rendering

65 factors XY RMSE 11.398 to 1.107 m mean |z| 6.129 to 0.693 m



no-BP rendering

12 factors XY RMSE 11.398 to 12.550 m mean |z| 6.129 to 1.261 m



— GT sonar mount

— baseline

— corrected

● accepted NAS-GS factor

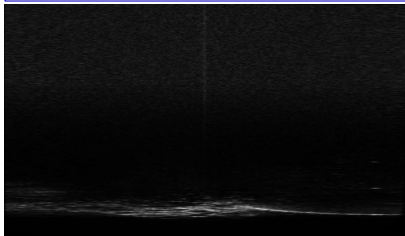
Offshore data contains stronger real-image artefacts, so BP and no-BP rendering are less interchangeable than in the DAVE Pass A replay.

Offshore Sonar Evidence

The real replay has stronger appearance mismatch than the DAVE route

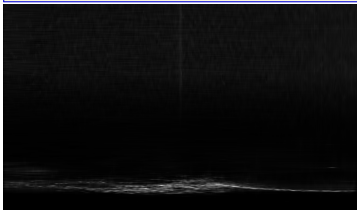
Recorded offshore

Live sonar from the replay.



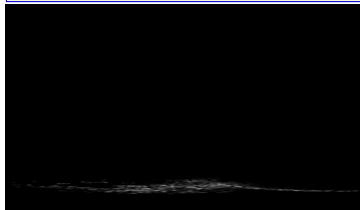
Rendered BP library

NAS-GS render with the offshore BP library.



Rendered no-BP library

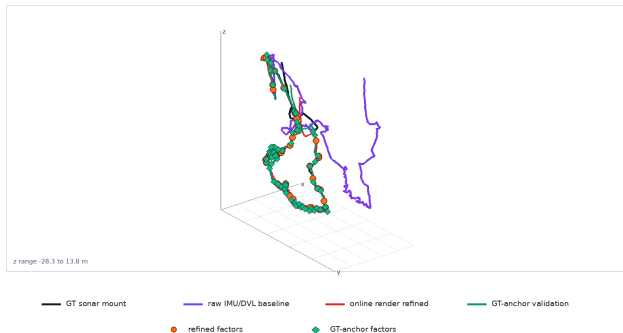
No-BP render used for appearance comparison.



Offshore Validation

The method transfers, but the real-scene case exposes z and appearance limits

Offshore 3D pose graph with live online-render correction



GT-anchor curve is the archived upper-bound diagnostic. Baseline and online-render are current live replay results.

Metrics: 50 factors. XY RMSE 11.043 $\rightarrow$ 1.407 m; XYZ RMSE 12.795 $\rightarrow$ 2.126 m; mean $|z|$ 6.074 $\rightarrow$ 1.027 m.

Final error: XY 9.843 m; $|z| = 3.601$ m.

Revisit Baselines

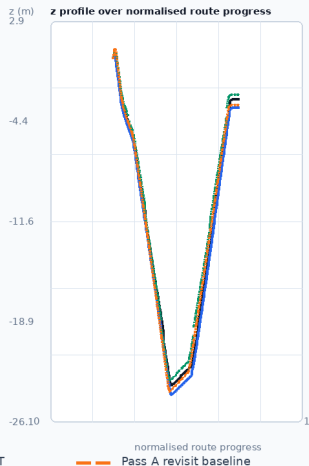
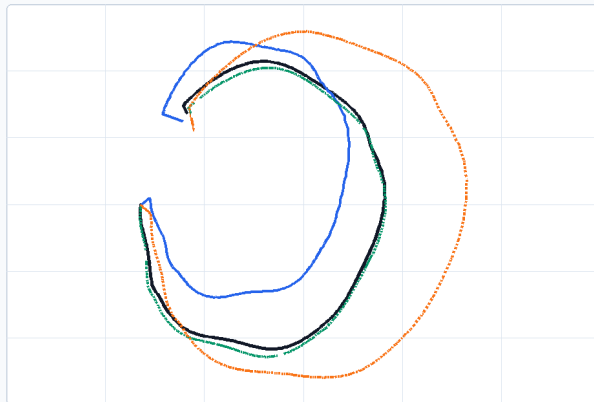
Pass A revisit is same-route; Pass B is harder because the route direction changes

Pass A and Pass A revisit realistic-drift baselines

Top-view uses the graph baseline. The z panel uses raw IMU/DVL odometry, where baseline spans 24.7 m and 24.8 m against GT spans 23.9 m and 23.9 m.

Pass A revisit is yaw-aligned to original Pass A for display only. Alignment yaw 70.1 deg.

top-view trajectory in online map frame



Revisit Result

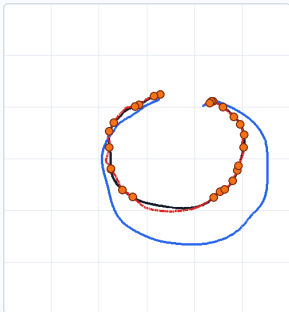
The revisit policy improves the hard case conservatively

Route-progress and same-route revisit replay results

Local single is strong for the same-route Pass A revisit. Route-progress gives guarded partial correction for the harder revisit cases.

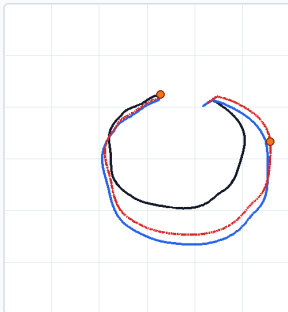
Pass A revisit local single

26 factors XY 2.901 to 1.008 m mean $|z|$ 12.123 to 1.034 m



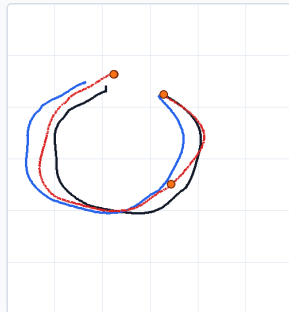
Pass A revisit route-progress

2 factors XY 2.901 to 2.442 m mean $|z|$ 12.123 to 1.946 m



Pass B route-progress

4 factors XY 2.734 to 1.875 m mean $|z|$ 13.115 to 3.605 m



— GT sonar mount

— baseline

- - - corrected

● accepted NAS-GS factor

Limitations and Future Work

The next steps follow directly from the observed failure modes

Current limitations

The remaining weaknesses define the operating boundary.

- The current system is evaluated through live replay and does not yet estimate a complete 6-DoF correction.
- The NAS-GS Gaussian map is fixed during localisation and cannot adapt to map or appearance changes.
- Route-progress registration provides only limited improvement under the most difficult revisit conditions.

Key future work

Each direction targets one demonstrated limitation.

- Move from live replay to onboard timing, calibration, and safety trials with 6-DoF estimation.
- Retrain NAS-GS on corrected trajectories and validate against an independent reference.
- Use sequence-level optimisation for robust large-drift recovery.
- Add differentiable sonar-intensity error for continuous pose refinement beyond discrete gates.

Main Contributions

- **Integrated simulation framework** DAVE data collection, NAS-GS training, and a live ROS Pose3 SLAM system.
- **Real-data-validated registration policies** Controlled IMU/DVL baselines and gated online-render factors for pose-graph correction.
- **Foundation for sonar-based 3DGS SLAM** A working connection from renderable sonar maps to incremental iSAM2 constraints.

IMPERIAL

Thank you

Questions?